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Research Statement

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My research is in econometrics, time series, forecasting, machine learning, and AI. I study the performance of algorithms and forecasts when models are estimated from limited data or applied to a different distribution. My current papers examine human–algorithm replacement, adversarial density forecasts, tail-risk recalibration, and forecast-combination weights. They ask when the gain from a more flexible method is large enough to justify the uncertainty involved in estimating and using it.

Human–algorithm replacement under distribution shift

My job market paper, *Robust Replacement of Human Decision Makers under Distribution Shift*, is a revise-and-resubmit at the *Journal of Econometrics*. It studies whether a fixed machine classification rule should replace a human when the machine's inputs may change at deployment. The analysis assumes that the human's class-conditional error rates remain stable.

A label-preserving Wasserstein neighborhood allows the machine's inputs to change. For each fixed threshold, I calculate the largest false-positive rate and smallest true-positive rate over that neighborhood. Together, they form a robust ROC frontier for comparison with the posterior for human error rates. Replacement requires strong enough posterior evidence that neither error rate will deteriorate. I bound the effects of frontier estimation on replacement decisions and welfare. The welfare bound retains a first-order threshold term; a margin condition yields a higher-order term for changes in the replacement set.

The numerical work illustrates how the choice changes with the permitted shift. One application uses 2018 American Community Survey income records, taking California, New York, and Texas as reference populations and other states as evaluation populations. Human decisions in this exercise are simulated, so the comparison isolates the effect of changing machine inputs.

Adversarial density forecasts in short samples

In *Adversarial Density Forecast for Macro-financial Risks*, Tae-Hwy Lee and I study conditional density forecasts estimated by Wasserstein generative methods. Flexible models can fit features of a distribution that parametric forecasts miss, but short macroeconomic samples make their estimates sensitive to training choices. We add a penalty that matches the variance and selected quantiles of generated outcomes to those in the observed data.

The theory connects this regularization to bounds on Wasserstein forecast error and expected continuous ranked probability score, allowing for finite simulation draws. In simulations, it improves performance over the unregularized adversarial estimator and reduces variation across fits. We also evaluate monthly U.S. unemployment, inflation, and industrial-production forecasts. The gains differ across series, with the clearest improvement over the unregularized method in unemployment forecasting.

Adjustment of joint tail-risk forecasts

Median-Anchored Adjustment of Joint VaR–ES Forecasts, also with Tae-Hwy Lee, is under review at the *Journal of Business & Economic Statistics*. It addresses a fitted risk model whose tail forecasts could be improved without re-estimating the whole model. We hold the fitted median fixed and multiply the distances to Value-at-Risk (VaR) and Expected Shortfall (ES) by one positive factor. The adjusted forecasts retain the correct VaR–ES ordering.

We derive a closed-form weighted-quantile estimator for the factor and study its large-sample properties. In rolling forecasts for the S&P 500, the adjustment is most useful when the baseline captures volatility dynamics but misses the shape of the tail.

Forecast combination and the cost of estimating weights

I study the choice between equal weights and estimated weights in *When Should Forecasts Be Equally Weighted? Signal Recovery and Estimation Costs*. I express the expected squared-error gain from estimated weights as recovered predictive signal minus the costs of estimating the component forecasts and combining weights. A tuning parameter selected from the data must also be treated as part of the forecasting procedure. Ignoring refitting after selection can even reverse the estimated gain.

For implementation, I compare the complete procedures using past out-of-sample forecast losses. Under stationarity and weak dependence, the selection rule favors the procedure with lower expected loss as the evaluation sample grows, provided the risks differ. The empirical analysis uses U.S. macroeconomic forecasts. In some settings, the simple average remains the better choice once weight estimation and tuning are included.

Future research

I plan to extend the human–algorithm comparison to settings in which human performance can also change with the environment. Updating replacement decisions as deployment data arrive would require estimating those changes while keeping track of the uncertainty that remains about both human and machine performance.

My forecasting work suggests a second direction: allowing a small amount of model flexibility to respond to observable economic conditions. This includes state-dependent VaR–ES adjustment factors, forecast-combination choices that vary with predictive conditions, and distributional anchors chosen for the forecast horizon or tail outcome. These extensions share an estimation problem: a method that reacts to changing conditions may improve forecasts, but only if there are enough observations to estimate that response reliably.